

CISC 121 – Assignment-3 and Template

Problem

Solve **one problem** from each of the five categories.

Stacks:

- Valid Parentheses ([LeetCode 20](#))
- Daily Temperatures ([LeetCode 739](#))

Queues:

- Number of Students Unable to Eat Lunch ([LeetCode 1700](#))
- Number of People Aware of a Secret ([LeetCode 2327](#))

Recursion

- Climbing Stairs ([LeetCode 70](#))
- Power of Three ([Leetcode 326](#))

Searching Problems:

- First Bad Version ([LeetCode 278](#))
- Find First and Last Position of Element in Sorted Array ([LeetCode 34](#))

Sorting Problems:

- Valid Anagram ([LeetCode 242](#))
- Sort Colors ([LeetCode](#))

✓ Problem 1

1. Understanding & Breakdown

*Understand the problem, inputs/outputs, constraints, and write down your plan or steps to solve the problem or draw the flowchart using digital or real pen-paper.
Insert the image.*

AI Level 1 is allowed to understand the concept, but you need to sketch the problem down by yourself.

Your effort is more important than the accuracy for this section

2. Peer Review & Draft Code

Review Guidance:

Visit the **"Solutions"** tab on LeetCode for the selected problem and review at least one top-rated solution. You may also search for relevant discussions or Q&A threads on **Stack Overflow** or other forums to understand how others approached the problem.

Then, write your own version based on what you understood**. The goal is not to copy but to apply what you learned. Always include links to the resources you reviewed.**

Example Answer (for "Palindrome Number"):

I reviewed the top solution under the LeetCode "Solutions" tab and one post on Stack Overflow:

<https://stackoverflow.com/questions/17129222/how-to-check-if-a-number-is-palindrome-in-python>

I learned that I can reverse the integer using a loop with modulo and division instead of converting it to a string. Using this idea, I tried writing a function that reverses the number and compares it to the original.

✓ 3. Final Correct Code

This version must pass all required test cases and handle edge cases correctly.

At this stage, you may:

- Use reviewed solutions or programming AI tools (e.g., ChatGPT, GitHub Copilot) to refine and fix errors.
- Refactor your draft to ensure accuracy and completeness.

AI Level 4 is allowed, but you **must include a disclaimer** that clearly states what tools you used, how you used them, and for which part of the code.

```
# Final working solution
# [AI Disclaimer example: Used ChatGPT with prompt "How to handle recursive bracket matching"]
```

✓ 4. Review & Fixes

What did you improve from your draft? 2 fixes with your comments.

```
# Fix 1: Simplified bracket matching using a dictionary

# Draft version (manual and repetitive logic)
if char == ")" and stack[-1] == "(": stack.pop()
elif char == "]" and stack[-1] == "[": stack.pop()
elif char == "}" and stack[-1] == "{": stack.pop()
else: return False

# Fixed version (cleaner and scalable)
mapping = {"(": ")", "[": "]", "{": "}"
if stack:
    top = stack.pop()
else:
    top = "#"
if mapping[char] != top:
    return False
```

```
# Fix 2: Prevented runtime error when stack is empty

# Draft version: crashes if the stack is empty
top = stack.pop() # This causes IndexError if stack is empty

# Fixed version: avoids crash by using fallback value
top = stack.pop() if stack else '#'
```

5. Reflection on Your Contribution

Where did AI or Other's code helped or fail to help? What specific logic or debugging did you contribute yourself?

Example

1. I used ChatGPT (Level 4) to help identify an edge case where my draft code failed when the input string started with a closing bracket. I also used ChatGPT to help structure the final loop and mapping dictionary, but verified all logic myself by tracing the code.
2. I used ChatGPT to help write the loop. But the code had an indentation error after the `if` block, which caused it to fail. I fixed the spacing and made sure the code runs correctly.
3. I used opensource code from Github. I found and fixed a logic bug where they used `if s[i] == '(' or ')'`, which always returned True. I corrected it to properly check both conditions.

✓ Repeat this template for Problems 2–5

You may copy-paste the above section 4 more times or duplicate cells in your notebook environment.

Remember: Proper attribution

No copy-pasting from AI or others

1. Your Name:
2. Section:
3. ID:
4. Github Link of this notebook:
5. Chat link with AI tool:

The most important part of writing code is not the code itself, but the thinking that leads to it.

Thank You